

DeepHANK

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The Model

We study a very very simple HANK model with:

- No capital
- No portfolio choice (firms' profits are mechanically redistributed to households)
- Perfectly inelastic labor supply
- No interest rate shocks
- Only Brownian TFP shocks

1.1 The Firms' Problem

1.1.1 The Final Good Producer

There is a continuum of competitive final good producers. Each final good producer uses intermediate goods Y_{jt} , $j \in [0, 1]$ as factors for the production of the final good. The production function is given as a CES aggregator with elasticity of substitution $\epsilon > 1$:

$$Y_t = \left(\int_0^1 Y_{jt}^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{1-\epsilon}}.$$

Letting P_t be the price of the final good and P_{jt} the price of the j th intermediate good, the representative final good producer's time t problem is

$$\begin{aligned} \max_{\{Y_{jt}\}_{j \in [0,1]}} & \left[P_t Y_t - \int_0^1 P_{jt} Y_{jt} dj \right] \\ \text{subject to} & \quad Y_t = \left(\int_0^1 Y_{jt}^{\frac{\epsilon-1}{\epsilon}} dj \right)^{\frac{\epsilon}{1-\epsilon}}. \end{aligned}$$

The first order condition for Y_{jt} is given as

$$P_t Y_t^{\frac{1}{\epsilon}} Y_{jt}^{-\frac{1}{\epsilon}} = P_{jt},$$

solving which we obtain the demand for the j th intermediate good:

$$Y_{jt} = \left(\frac{P_{jt}}{P_t} \right)^{-\epsilon} Y_t.$$

Substituting this into the CES aggregator shows us that

$$Y_t = Y_t P_t^\epsilon \left(\int_0^1 P_{jt}^{1-\epsilon} dj \right)^{\frac{\epsilon}{1-\epsilon}},$$

so that

$$P_t = \left(\int_0^1 P_{jt}^{1-\epsilon} dj \right)^{\frac{1}{1-\epsilon}}.$$

1.1.2 The Intermediate Good Producer

For each intermediate good there is a single firm that produces that good, so that the intermediate good producers monopolistically supply differentiated goods. They face the linear production function

$$Y_{jt} = Z_t n_{jt},$$

where n_{jt} is the amount of labor producer j uses and Z_t is TFP. We assume that $\log Z_t = z_t$ follows a diffusion process

$$dz_t = \mu_{zt} dt + \sigma_z dW_t^0,$$

where $\{W_t^0\}_{t \in \mathbb{N}}$ is a standard Wiener process.

Let π_{jt} be the time t inflation rate of the price of the j th intermediate good, so that P_{jt} evolves according to the process

$$dP_{jt} = \pi_{jt} P_{jt} dt.$$

We assume that producer j faces the Rotemberg adjustment cost

$$\Psi_t(\pi_{jt}) = \frac{\psi}{2} (\pi_{jt})^2 Y_t,$$

which in terms of real output. In other words, producer j incurs a menu cost if it adjusts the price in any way.

Let r_t be the real interest rate and w_t the real wage. Now producer j 's faces the intertemporal problem

$$\max_{\{\pi_{jt}\}_{t \in \mathbb{N}}} \mathbb{E}_0 \left[\int_0^\infty e^{-\int_0^t r_s ds} (\Pi_t(P_{jt}) - \Psi_t(\pi_{jt})) dt \right]$$

$$\text{subject to } dP_{jt} = \pi_{jt} P_{jt} dt$$

$$\Pi_t(P_{jt}) = \left(\frac{P_{jt}}{P_t} \right)^{1-\epsilon} Y_t - \frac{w_t}{e^{z_t}} \left(\frac{P_{jt}}{P_t} \right)^{-\epsilon} Y_t$$

$$\Psi_t(\pi_{jt}) = \frac{\psi}{2} (\pi_{jt})^2 Y_t$$

where everything is in terms of real output. The corresponding HJBE is

$$r_t W_t(P^j) = \max_{\pi^j} \left[\left(\frac{P^j}{P_t} \right)^{1-\epsilon} Y_t - \frac{w_t}{e^{z_t}} \left(\frac{P^j}{P_t} \right)^{-\epsilon} Y_t - \frac{\psi}{2} (\pi^j)^2 Y_t + \partial_{P^j} W_t(P^j) \pi^j P^j \right] + \partial_t W_t(P^j),$$

where we suppress the dependence of $W_t(P^j)$ on z_t, r_t, w_t, P_t, Y_t .

The equilibrium path of $\{\pi_{jt}, P_{jt}\}_{t \in \mathbb{N}}$ satisfies the first order condition

$$\psi \pi_{jt} Y_t = \partial_{P^j} W_t(P_{jt}) P_{jt},$$

while the envelope condition with respect to P^j tells us that

$$\begin{aligned} r_t \partial_{P^j} W_t(P_{jt}) &= \left((1 - \epsilon) + \epsilon \frac{w_t}{p_{jt} e^{z_t}} \right) p_{jt}^{-\epsilon} \frac{Y_t}{P_t} \\ &\quad + \partial_{P^j P^j} W_t(P_{jt}) \pi_{jt} P_{jt} + \partial_{P^j} W_t(P_{jt}) \pi_{jt} \\ &\quad + \partial_{P^j t} W_t(P_{jt}), \end{aligned}$$

where $p_{jt} = \frac{P_{jt}}{P_t}$ is the relative price of good j . Using the first order condition and multiplying both sides by $\frac{1}{\psi} \frac{P_{jt}}{Y_t}$, we can rewrite the envelope condition as

$$\begin{aligned} r_t \pi_{jt} &= \frac{\epsilon}{\psi} \left(\frac{1 - \epsilon}{\epsilon} + \frac{w_t}{p_{jt} e^{z_t}} \right) p_{jt}^{1-\epsilon} \\ &\quad + \frac{1}{\psi} \left[\left(\partial_{P^j P^j} W_t(P_{jt}) \frac{P_{jt}}{Y_t} + \partial_{P^j} W_t(P_{jt}) \frac{1}{Y_t} \right) \pi_{jt} P_{jt} + \partial_{P^j t} W_t(P_{jt}) \frac{P_{jt}}{Y_t} \right] \end{aligned}$$

To obtain the diffusion for π_{jt} , note from the first order condition that

$$\pi_{jt} = \frac{1}{\psi} \partial_{P^j} W_t(P_{jt}) \frac{P_{jt}}{Y_t}.$$

Defining the mapping $f(t, P^j, Y)$ as

$$f(t, P^j, Y) = \frac{1}{\psi} \partial_{P^j} W_t(P^j) \frac{P^j}{Y},$$

Ito's lemma tells us that

$$\begin{aligned} d\pi_{jt} &= \frac{\partial f(t, P_{jt}, Y_t)}{\partial t} dt + \frac{\partial f(t, P_{jt}, Y_t)}{\partial P^j} dP_{jt} + \frac{\partial f(t, P_{jt}, Y_t)}{\partial Y} dY_t + \frac{1}{2} \frac{\partial^2 f(t, P_{jt}, Y_t)}{\partial Y^2} (dY_t)^2 \\ &= \frac{1}{\psi} \left[\partial_{P^j t} W_t(P_{jt}) \frac{P_{jt}}{Y_t} + \left(\partial_{P^j P^j} W_t(P_{jt}) \frac{P_{jt}}{Y_t} + \partial_{P^j} W_t(P_{jt}) \frac{1}{Y_t} \right) \pi_{jt} P_{jt} \right] dt \\ &\quad - \frac{1}{\psi} \partial_{P^j} W_t(P_{jt}) \frac{P_{jt}}{Y_t} \left(\frac{dY_t}{Y_t} \right) + \frac{1}{\psi} \partial_{P^j} W_t(P_{jt}) \frac{P_{jt}}{Y_t} \left(\frac{dY_t}{Y_t} \right)^2, \end{aligned}$$

where there are no second order terms involving dP_{jt} because it consists only of a drift term.

Now the rewritten envelope condition immediately tells us that the drift term above can be

expressed as

$$r_t \pi_{jt} - \frac{\epsilon}{\psi} \left(\frac{1-\epsilon}{\epsilon} + \frac{w_t}{p_{jt} e^{z_t}} \right) p_{jt}^{1-\epsilon},$$

so that π_{jt} evolves according to

$$d\pi_{jt} = \left[r_t \pi_{jt} - \frac{\epsilon}{\psi} \left(\frac{1-\epsilon}{\epsilon} + \frac{w_t}{p_{jt} e^{z_t}} \right) p_{jt}^{1-\epsilon} \right] dt + \pi_{jt} \left[\left(\frac{dY_t}{Y_t} \right)^2 - \frac{dY_t}{Y_t} \right].$$

1.2 The Household's Problem

There is a continuum of households indexed by $i \in [0, 1]$ that chooses how much to consume each period and saves in a bond in zero net supply subject to a soft borrowing penalty. We abstract from two aspects of the household problem: labor and portfolio choice. For labor supply, we assume that households face uninsurable shocks to employment status in the form of an exogenous Poisson process, while we abstract from the portfolio choice problem by assuming that each household receives a fixed and equal share of each intermediate good producer's profits each period.

Suppose that household i believes that the time t interest rate, wage and profits are given as $\tilde{r}_t$, $\tilde{w}_t$ and $\tilde{\Pi}_t$. Under these beliefs, household i 's problem can be written as

$$\begin{aligned} \max_{\{c_{it}\}_{t \in \mathbb{N}}} \quad & \mathbb{E}_0 \left[\int_0^\infty e^{-\rho t} \left(u(c_{it}) + \varphi(a) \cdot I_{\{a \leq \underline{a}\}} \right) dt \right] \\ \text{subject to} \quad & d \begin{bmatrix} a_{it} \\ n_{it} \end{bmatrix} = \begin{bmatrix} \tilde{r}_t a_{it} + \tilde{w}_t n_{it} - c_{it} + \tilde{\Pi}_t \\ 0 \end{bmatrix} dt + \begin{bmatrix} 0 \\ \check{n}_{it} - n_{it} \end{bmatrix} dN_t^i \end{aligned}$$

where $\{N_t^i\}_{t \geq 0}$ is a Poisson process with intensity $\lambda(N_t^i)$ that is independent across households and $\varphi(a) \cdot I_{\{a \leq \underline{a}\}}$ penalizes savings below $\underline{a}$. The corresponding HJBE of the household is given as

$$\begin{aligned} \rho V_t(a, n) = \max_{c \geq 0} \quad & \left[u(c) + \varphi(a) \cdot I_{\{a \leq \underline{a}\}} + \partial_a V_t(a, n) (\tilde{r}_t a + \tilde{w}_t n - c + \tilde{\Pi}_t) \right. \\ & \left. + \lambda(n) (V_t(a, \check{n}) - V_t(a, n)) \right] + \partial_t V_t(a, n) \end{aligned}$$

where the dependence of $V_t(\cdot)$ on the beliefs $\tilde{r}_t, \tilde{w}_t, \tilde{\Pi}_t$ is suppressed. The first order condition for consumption is, as usual, given as

$$u'(c_t(a, n)) = \partial_a V_t(a, n)$$

for $a > \underline{a}$. Finally, we denote the density of the wealth and employment status distribution at time t by $g_t(\cdot)$.

Since the evolution of a_{it} does not depend directly on the aggregate source of risk W_t^0 , it follows that the KFE is given as

$$\begin{aligned} \partial_t g_t(a, n \mid \tilde{r}_t, \tilde{w}_t, \tilde{\Pi}_t) = -\partial_a \left[\left(\tilde{r}_t a + \tilde{w}_t n - c_t(a, n) + \tilde{\Pi}_t \right) g_t(a, n \mid \tilde{r}_t, \tilde{w}_t, \tilde{\Pi}_t) \right] \\ + \lambda(\check{n}) g_t(a, \check{n} \mid \tilde{r}_t, \tilde{w}_t, \tilde{\Pi}_t) - \lambda(n) g_t(a, n \mid \tilde{r}_t, \tilde{w}_t, \tilde{\Pi}_t). \end{aligned}$$

1.3 The Interest Rate Rule and Equilibrium

1.3.1 The Interest Rate Rule

We assume that the central bank sets the nominal interest rate according to the rule

$$i_t = r^* + \phi_\pi \pi_t,$$

where there are no interest rate shocks at the moment. The model is closed by the Fisher equation $i_t = r_t + \pi_t$, so that the interest rate r_t is determined as

$$r_t = r^* + (\phi_\pi - 1)\pi_t.$$

Under the active monetary policy assumption $\phi_\pi > 1$, this equation implies that the real interest rate moves in the same direction as inflation π_t .

1.3.2 Rational Expectations Equilibrium

In a symmetric equilibrium, P_{jt} is equal across j , so that $P_{jt} = P_t$ and every intermediate good producer demands the same amount of labor. By implication, $Y_{jt} = Y_t$ for any j and t , and aggregate labor demand is given by

$$N_t^d = \int_0^1 n_{jt}^d dj = \frac{Y_t}{e^{z_t}}.$$

Furthermore, the inflation rate π_t corresponding to the price P_t of the final good is the same as the inflation rate π_{jt} governing the evolution of the price P_{jt} of the j th intermediate good, and we have the usual NKPC

$$d\pi_t = \left[r_t \pi_t + \frac{\epsilon}{\psi} \left(\frac{\epsilon - 1}{\epsilon} - \frac{w_t}{e^{z_t}} \right) \right] dt + \pi_t \left[\left(\frac{dY_t}{Y_t} \right)^2 - \frac{dY_t}{Y_t} \right],$$

which in light of the interest rate rule can alternatively be written as

$$d\pi_t = \left[r^* \pi_t + (\phi_\pi - 1)\pi_t^2 + \frac{\epsilon}{\psi} \left(\frac{\epsilon - 1}{\epsilon} - \frac{w_t}{e^{z_t}} \right) \right] dt + \pi_t \left[\left(\frac{dY_t}{Y_t} \right)^2 - \frac{dY_t}{Y_t} \right].$$

Finally, the real profits ex-adjustment costs of the intermediate good producers are all equal to

$$\Pi_t = \left(1 - \frac{w_t}{e^{z_t}} - \frac{\psi}{2} \pi_t^2 \right) Y_t.$$

As noted, this is always non-negative.

A symmetric REE in this model is a collection of stochastic processes $\{c_t(\cdot)g_t(\cdot), P_t, Y_t, q_t = (\mu_{rt}, \sigma_{rt}, w_t, \Pi_t)\}_{i \in [0,1], t \in \mathbb{N}}$ such that:

- 1) Given the belief $\tilde{q}_t = (\tilde{\mu}_{rt}, \tilde{\sigma}_{rt}, \tilde{w}_t, \tilde{\Pi}_t)$ about the price processes, $\{c_t(\cdot)\}_{t \in \mathbb{N}}$ is a sequence of consumption policy functions that solve the household's problem, and $g_t(\cdot)$ is the resulting distribution of wealth and labor supply at time t .

- 2) The intermediate good producers' optimality condition (the NKPC) is satisfied:

$$d\pi_t = \left[r^* \pi_t + (\phi_\pi - 1) \pi_t^2 + \frac{\epsilon}{\psi} \left(\frac{\epsilon - 1}{\epsilon} - \frac{w_t}{e^{z_t}} \right) \right] dt + \pi_t \left[\left(\frac{dY_t}{Y_t} \right)^2 - \frac{dY_t}{Y_t} \right].$$

- 3) The market clearing conditions are satisfied for any $t \geq 0$:

$$\frac{Y_t}{e^{z_t}} = N^* := \sum_{j=1}^2 n_j \int_0^1 g_t(a, n_j) da \quad (\text{Labor Market})$$

$$dB_t = r_t B_t dt, \quad B_0 = 0 \quad \Leftrightarrow \quad B_t = 0 \quad \forall t \geq 0 \quad (\text{Bond Market})$$

$$Y_t = \sum_{j=1}^2 \int_{-\infty}^{\infty} c_t(a, n) g_t(a, n) da + \frac{\psi}{2} \pi_t^2 Y_t \quad (\text{Goods Market})$$

where $B_t = \sum_{j=1}^2 \int_{-\infty}^{\infty} a g_t(a, n_j) da$

- 4) The firms earn zero overall profits:

$$Y_t = w_t N^* + \frac{\psi}{2} \pi_t^2 Y_t$$

- 5) The real interest rate is determined by the interest rate rule:

$$r_t = r^* + (\phi_\pi - 1) \pi_t.$$

- 6) Beliefs are consistent, so that $\tilde{q}_t = q_t$.

1.4 The Master Equation

In symmetric REE, $Y_t = e^{z_t} N^*$, so by Ito's lemma

$$\frac{dY_t}{Y_t} = \left(\mu_z(z_t) + \frac{1}{2} \sigma_z^2 \right) dt + \sigma_z dW_t^0$$

$$\left(\frac{dY_t}{Y_t} \right)^2 = \sigma_z^2 dt,$$

so that the NKPC can be written as

$$d\pi_t = \left[r^* \pi_t + (\phi_\pi - 1) \pi_t^2 + \frac{\epsilon}{\psi} \left(\frac{\epsilon - 1}{\epsilon} - \frac{w_t}{e^{z_t}} \right) - \left(\mu_z(z_t) - \frac{1}{2} \sigma_z^2 \right) \pi_t \right] dt - \sigma_z \pi_t dW_t^0.$$

In this context belief consistency requires

$$\tilde{r}_t = r_t = r^* + \phi_\pi \pi_t \quad \tilde{w}_t = w_t, \quad \tilde{\Pi}_t = \Pi_t.$$

Finally, the wage each period can be obtained from the bond market clearing condition as a function of the aggregate states, which are log TFP z_t , inflation π_t and the wealth distribution $g_t(\cdot)$.

Now equilibrium in this economy can be characterized by one master equation. In the fully recursive formulation of the household's problem, the household's value function depends on the individual states a, n and the aggregate stats z, π, g . The state variables are known to evolve according to

$$\begin{aligned} da_t &= s(a_t, n_t, z_t, \pi_t, g_t) dt \\ dn_t &= (\check{n}_t - n_t) dN_t \\ dz_t &= \mu_z(z_t) dt + \sigma_z dW_t^0 \\ d\pi_t &= \mu_\pi(z_t, \pi_t, g_t) dt - \sigma_z \pi_t dW_t^0 \\ dg_t(a, n) &= \mu_g(a, n; z_t, \pi_t, g_t) dt \end{aligned}$$

where

$$\begin{aligned} s(a, n, z, \pi, g) &= r(z, \pi, g) a + w(z, \pi, g) n - (u')^{-1} [\partial_a V(a, n, z, \pi, g)] \\ \mu_\pi(z, \pi, g) &= r^* \pi + (\phi_\pi - 1) \pi^2 + \frac{\epsilon}{\psi} \left(\frac{\epsilon - 1}{\epsilon} - \frac{w(z, \pi, g)}{e^z} \right) - \left(\mu_z(z) - \frac{1}{2} \sigma_z^2 \right) \pi \\ \mu_g(a, n; z, \pi, g) &= -\partial_a (s(a, n, z, \pi, g) g(a, n)) + \lambda(\check{n}) g(a, \check{n}) - \lambda(n) g(a, n) \\ w(z, \pi, g) &= \left(1 - \frac{\psi}{2} \pi^2 \right) e^z \end{aligned}$$

$$r(z, \pi, g) = r^* + \phi_\pi \pi$$

Therefore, the master equation is given as

$$\begin{aligned}
 \rho V(a, n, z, \pi, g) = & u \left((u')^{-1} [\partial_a V(a, n, z, \pi, g)] \right) & (\text{Flow Utility} + \text{FOC}) \\
 & + \varphi'(a) \cdot I_{\{a \leq \underline{a}\}} & (\text{Borrowing Penalty}) \\
 & + \partial_a V(a, n, z, \pi, g) s(a, n, z, \pi, g) & (\text{Individual Wealth}) \\
 & + \lambda(n) \left(V(a, \check{n}, z, \pi, g) - V(a, n, z, \pi, g) \right) & (\text{Individual Labor Supply}) \\
 & + \mu_z(z) \partial_z V(a, n, z, \pi, g) + \frac{1}{2} \sigma_z^2 \partial_{zz} V(a, n, z, \pi, g) & (\text{TFP}) \\
 & + \mu_\pi(z, \pi, g) \partial_\pi V(a, n, z, \pi, g) + \frac{1}{2} \sigma_z^2 \pi^2 \partial_{\pi\pi} V(a, n, z, \pi, g) & (\text{Inflation}) \\
 & - \sigma_z^2 \pi \partial_{\pi z} V(a, n, z, \pi, g) & (\text{Cross Variation Term}) \\
 & + \sum_{j=1}^2 \int_{-\infty}^{\infty} D_g V(a', n_j; a, n, z, \pi, g) \mu_g(a', n_j; z, \pi, g) da' & (\text{Wealth Distribution})
 \end{aligned}$$

where

$$(a', n_j) \mapsto D_g V(a', n_j; a, n, z, \pi, g)$$

is the Frechet derivative of $V(a, n, z, \pi, g)$ with respect to the distribution g evaluated at the point (a', n_j) on the state space.

1.4.1 Finite Agent Approximation

We now approximate the wealth distribution using a finite number of agents. The approximate state space consists of a, n , the household's individual states, z, π , the finite dimensional macro aggregates, and $\varphi^{-0} = \{a_i, n_i\}_{1 \leq i \leq N}$, the wealth and labor supply positions of the other agents in the economy. Given our initial condition $B_0 = 0$, we choose a_N so that

$$\frac{1}{N} \sum_{i=1}^N a_i = 0.$$